Constructed with simple fixed-wing designs and lightweight materials, these drones operated solely through manual radio control, without the aid of sensors, GPS, or auto-mation. As shown in the figure, the design emphasizes basic aerodynamics and mechanical simplicity, reflecting the technological limitations of the time while laying the groundwork for future innovations in autonomous and intelligent UAV systems.

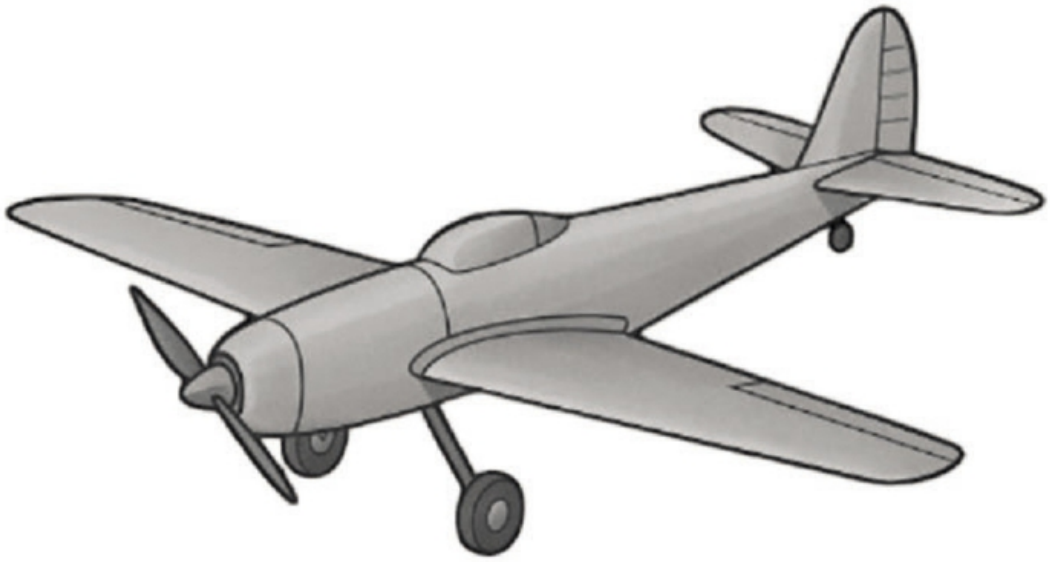


Figure 1.3 Manual drone (first generation UAV) ⏎

1.4.2 Second generation: Remote control and GPS integration

The second generation of drones marked a transformative period in the evolution of unmanned aerial systems, moving from basic manual operation to semi-automated and more reliable flight control. Emerging prominently during the 1980s and 1990s, this era witnessed significant technological improvements, primarily through the introduction of remote control systems, onboard electronics, and global positioning system (GPS) technology. These advancements collectively enhanced flight precision, stability, and mission endurance, expanding drone applications beyond military training into surveillance, research, and civilian domains.